

/CSCI 1101 WEB DEV

<HTML>



/WEB ACCESS



HOW THE WEB WORKS?

Local Device

> Computer, smart phone, IoT devices

DNS Servers

> Domain Name System servers = phone book

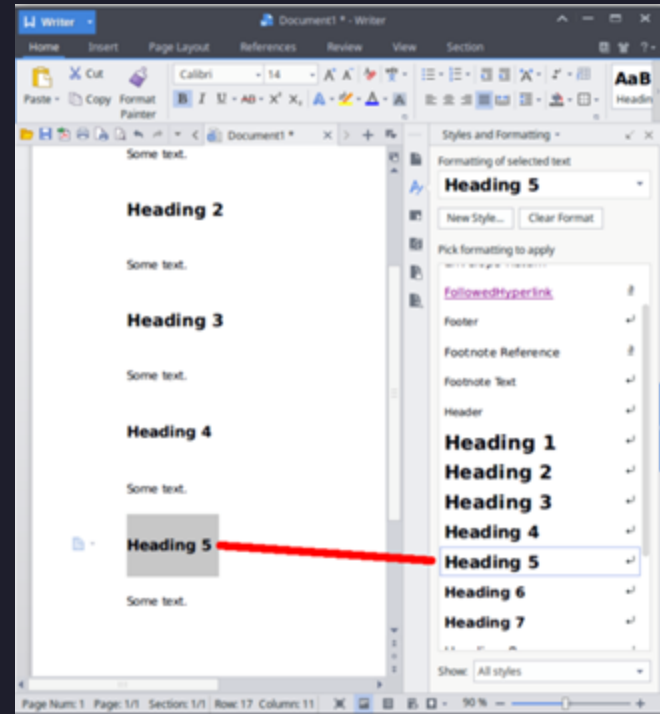
IP Address

> Internet Protocol (IP) Address = unique telephone number



/STRUCTURE

- Webpages are analogous to how Word docs are structured.
- The use of headings and subheadings often reflects a hierarchy of information.
- Structure facilitate the purpose and intention of the document.



/STRUCTURE OF HTML

`<html>`

`<body>`

`<h1>`My First Heading`</h1>`

`<p>`My first paragraph.`</p>`

`<h2>`This is a subheading`</h2>`

`<p>`Another paragraph.`</p>`

`</body>`

`</html>`

HTML tags (markup):



Document



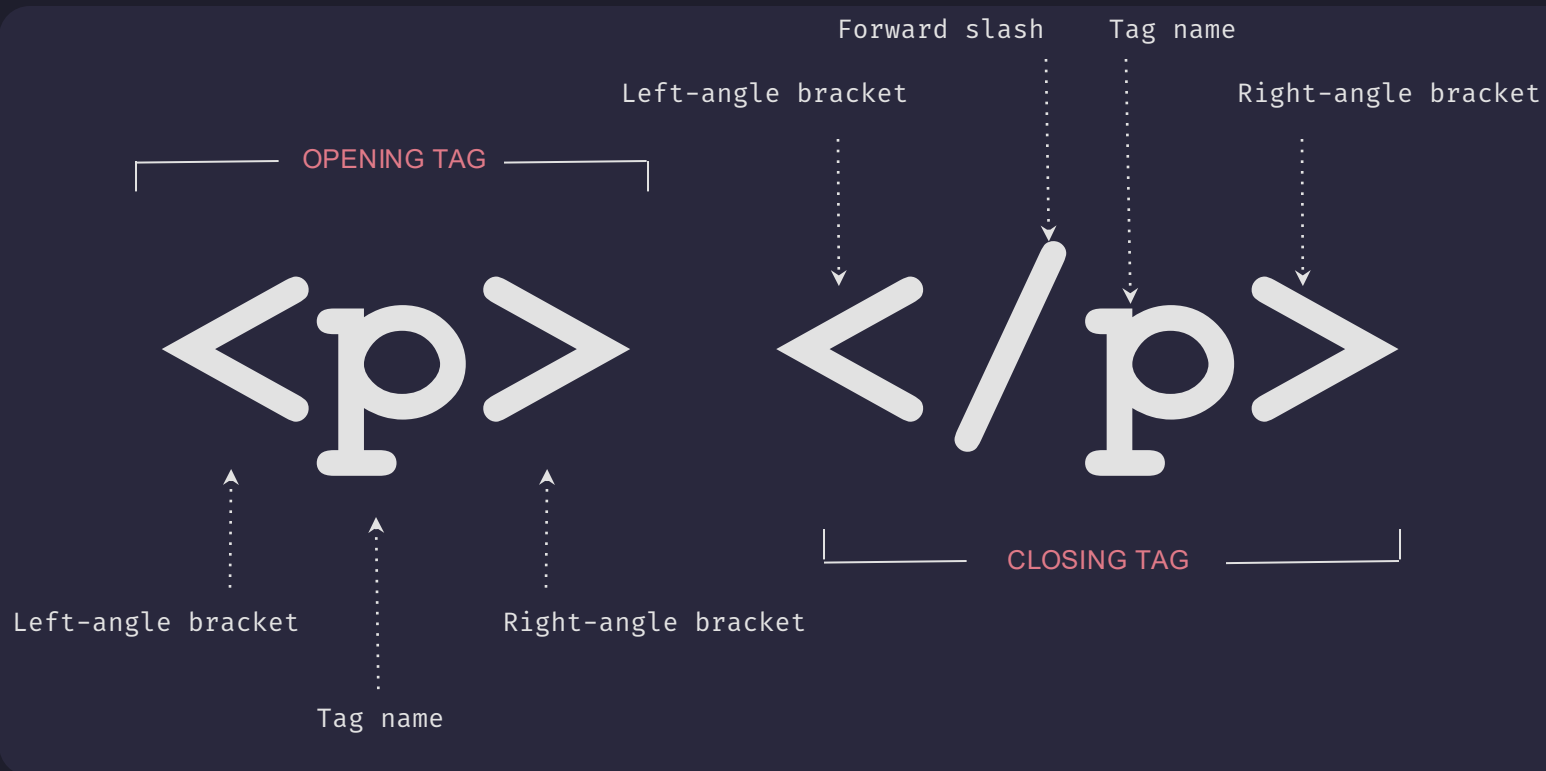
Headings



Paragraphs

Let's create our first web page!

Visual Studio 2022



>

ATTRIBUTES

<

THIS IS THE
ATTRIBUTE
NAME



```
<p lang="en-us">Paragraph in English</p>
```



THIS IS THE
ATTRIBUTE
VALUE

/TEXT

Heading and paragraphs, basic
styles, semantic markup



/HEADINGS



HTML

```
<h1>This is heading 1</h1>  
<h2>This is heading 2</h2>  
<h3>This is heading 3</h3>  
<h4>This is heading 4</h4>  
<h5>This is heading 5</h5>  
<h6>This is heading 6</h6>
```

RESULT



/PARAGRAPHS

HTML

```
<p>Better to spend your time  
at funerals than at parties.  
After all, everyone dies—so  
the living should take this  
to heart.</p>
```

```
<p>Sorrow is better than  
laughter,for sadness has a  
refining influence on us. A  
wise person thinks a lot  
about death,while a fool  
thinks only about having a  
good time. </p>
```

RESULT

Better to spend your time at funerals than at parties.
After all, everyone dies—so the living should take
this to heart.

Sorrow is better than laughter,for sadness has a
refining influence on us. A wise person thinks a lot
about death,while a fool thinks only about having a
good time.

/BOLD & ITALIC

HTML

```
<p>Better to spend your time  
at funerals than at parties.  
After all, everyone dies—so  
<b>the living should take  
this to heart</b>.</p>
```

```
<p><i>Sorrow</i> is better  
than laughter,for sadness  
has a refining influence on  
us. A <i>wise</i> person  
thinks a lot about  
death,while a fool thinks  
only about having a good  
time. </p>
```

RESULT

Better to spend your time at funerals than at parties.
After all, everyone dies—so **the living should take this
to heart.**

Sorrow is better than laughter,for sadness has a refining
influence on us. A *wise* person thinks a lot about
death,while a fool thinks only about having a good time.

/WHITES SPACE (white space collapsing)



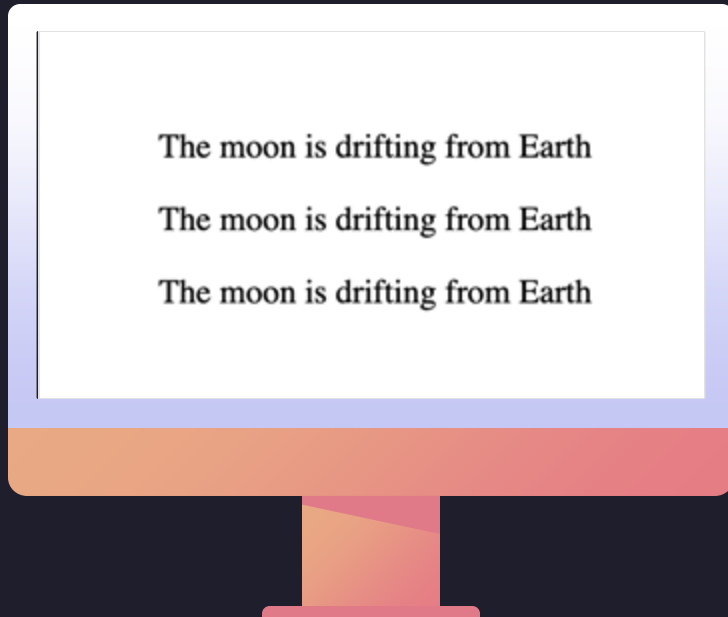
HTML

```
<p>The moon is drifting from  
Earth</p>
```

```
<p>The moon           is  
drifting from Earth</p>
```

```
<p>The moon is drifting from  
  
Earth</p>
```

RESULT



/STRONG & EMPHASIS

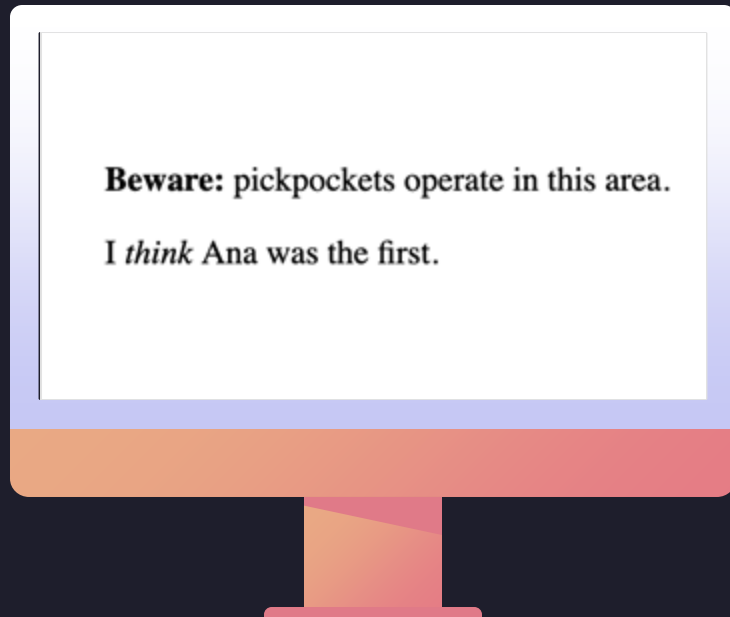


HTML

```
<p><strong> Beware: </strong>  
pickpockets operate in this  
area.</p>
```

```
<p>I <em> think </em> Ana was  
the first.</p>
```

RESULT



/LISTS

Numbered, bullet, and definition lists



/HTML Lists



/ORDERED

- > Each item in the list is numbered

/UNORDERED

- > A list that use bullets

/DEFINITION

- > These are made up of a set of terms along with the definitions for each of those terms

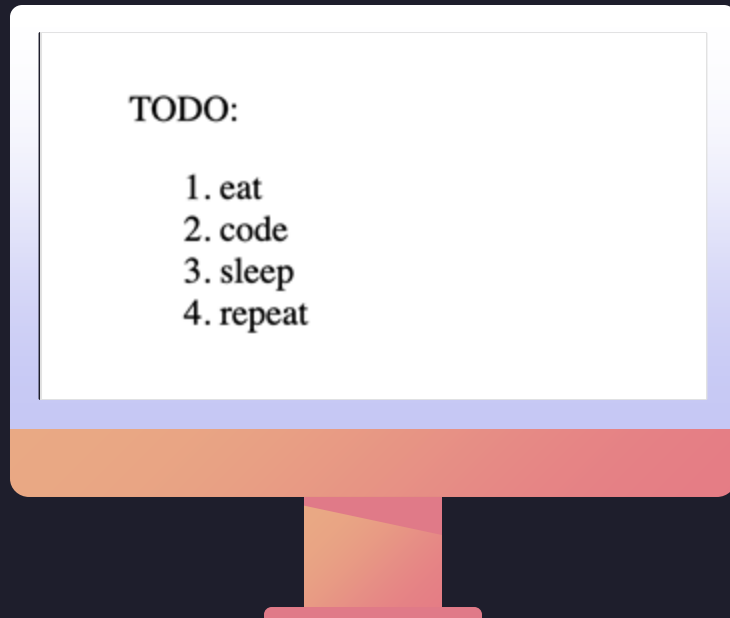
/ORDERED LISTS



HTML

```
<p>TODO:</p>
<ol>
  <li>eat </li>
  <li>code</li>
  <li>sleep</li>
  <li>repeat</li>
</ol>
```

RESULT



/UNORDERED LISTS

HTML

```
<p>Homemade Pasta Recipe Ingredients</p>
<ul>
  <li>All-purpose flour – Regular
  all-purpose flour yields chewy,
  bouncy noodles every time.</li>
  <li>Eggs – The key ingredient for
  adding richness and moisture to the
  dough!</li>
  <li>Olive oil – Along with the
  eggs, a splash of olive oil
  moistens the dough and helps it
  come together.</li>
  <li>Salt – Add it to the dough and
  the pasta water for the best
  flavor.</li>
</ul>
```

RESULT

Homemade Pasta Recipe Ingredients

- All-purpose flour – Regular all-purpose flour yields chewy, bouncy noodles every time.
- Eggs – The key ingredient for adding richness and moisture to the dough!
- Olive oil – Along with the eggs, a splash of olive oil moistens the dough and helps it come together.
- Salt – Add it to the dough and the pasta water for the best flavor.

/DEFINITION LISTS

HTML

```
<dl>
  <dt>Sashimi</dt>
  <dd>a Japanese dish of bite-
    sized pieces of raw fish eaten
    with soy sauce and wasabi
    paste.</dd>
  <dt>Scale</dt>
  <dd>an instrument for weighing,
    originally a simple balance ( a
    pair of scales ) but now usually
    a device with an electronic or
    other internal weighing
    mechanism.</dd>
</dl>
```

RESULT

Sashimi

a Japanese dish of bite-sized pieces of raw fish eaten with soy sauce and wasabi paste.

Scale

an instrument for weighing, originally a simple balance (a pair of scales) but now usually a device with an electronic or other internal weighing mechanism.

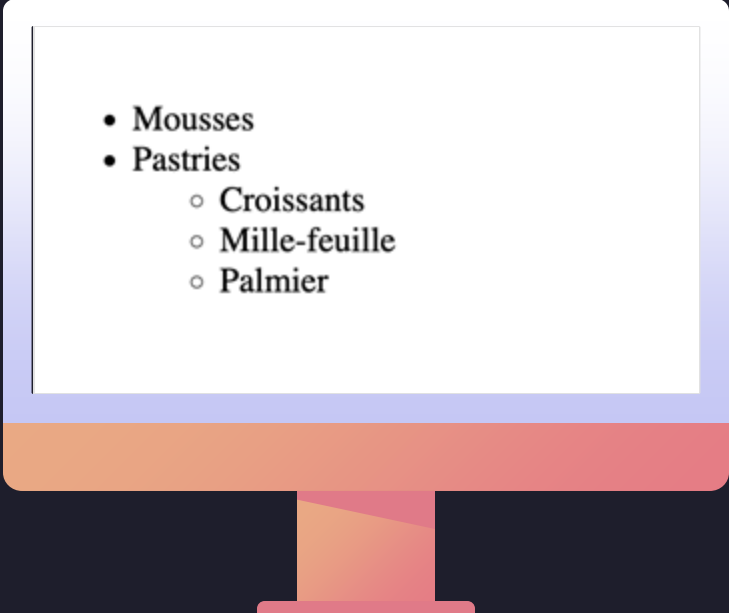
/NESTED LISTS



HTML

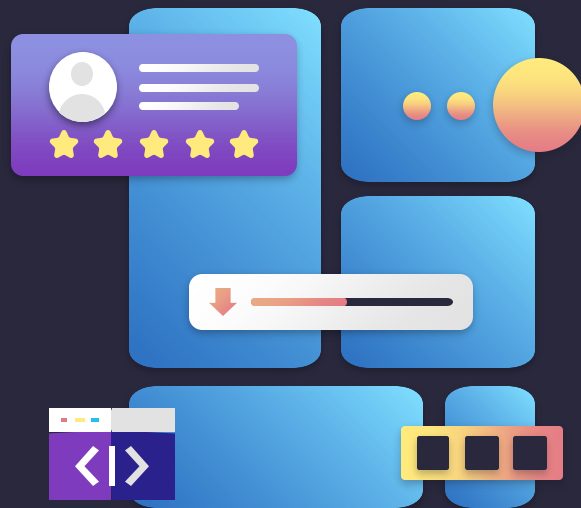
```
<ul>
  <li>Mousses</li>
  <li>Pastries</li>
  <ul>
    <li>Croissants</li>
    <li>Mille-feuille</li>
    <li>Palmier</li>
  </ul>
</ul>
```

RESULT

- 
- Mousses
 - Pastries
 - Croissants
 - Mille-feuille
 - Palmier

/LINKS

Creating links between pages, to
other sites and email links



/LINKS

We will learn about the different types of links:

- Links from one website to another
- Links from one page to another on the same website
- Links from one section of a page to another section of the same page
- Links to open in a new browser
- Links that start up your email program



WRITING LINKS



OPENING TAG

THIS IS THE
TEXT USERS
CAN CLICK
ON

```
<a href="https://ap-lab.ca">Applied Perception</a>
```

THIS IS THE
PAGE THE
LINK TAKES
USERS TO

CLOSING TAG

> OPENING LINKS IN A NEW WINDOW <

```
<a href="https://ap-lab.ca" target="_blank">AP Lab</a>
```

THIS OPENS
THE PAGE
SPECIFIED IN
THE HREF IN
A NEW
WINDOW

<URLs>

/UNIFORM RESOURCE LOCATOR

/ABSOLUTE URLs

- Absolute URLs start with the domain name for that site, and can be followed by the path to a specific page.

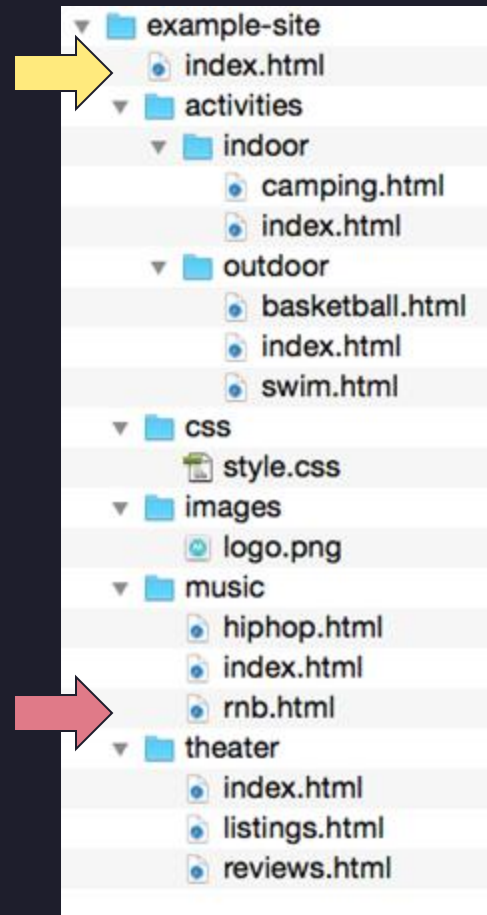
`<a href="www.example-site.com/music/rnb.html"> RNB </a>`

Diagram illustrating the structure of the href attribute:

- `www.example-site.com` is the **DOMAIN NAME**.
- `/music/rnb.html` is the **PATH TO PAGE**.

- If no page is specified, the site will display the contents of the `homepage` in the root, called `index.html`

```
<a href="www.example-site.com"> HOMEPAGE </a>
```

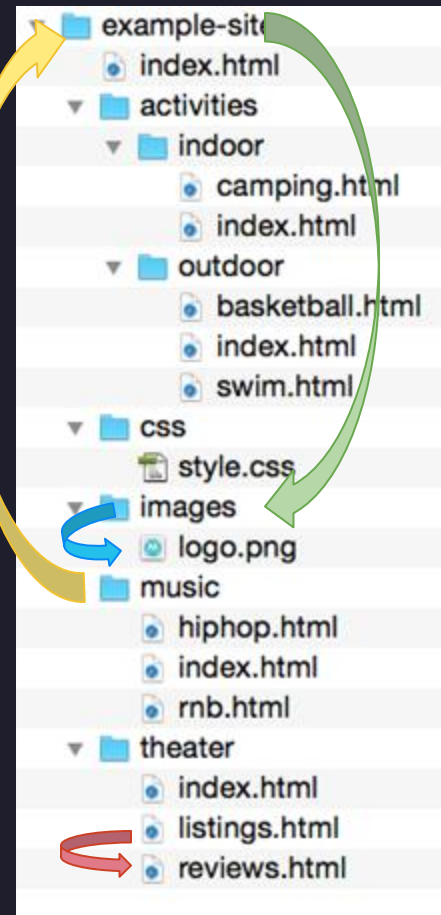


/RELATIVE URLs

Relative URLs do NOT include the domain name for the site.

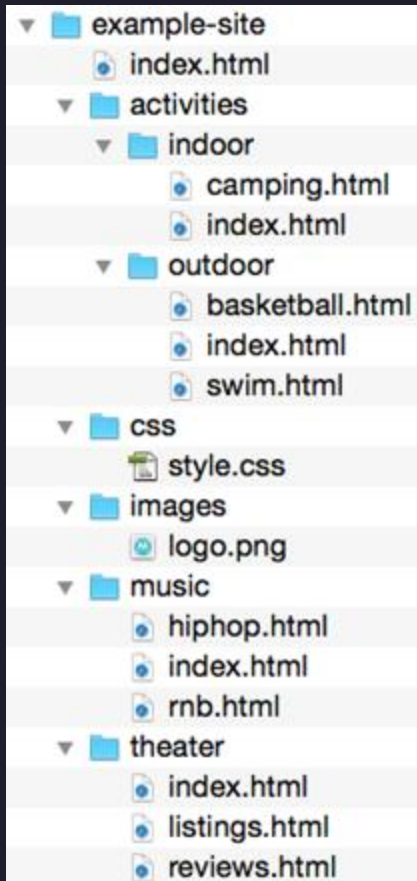
- To access pages within the same folder, we use the filename. For example, a link in the listing.html accessing the reviews page:
`<a href="reviews.html"> Reviews </a>`
- To access pages in different folders, we traverse up the directory by levels to access a specific resource. For example, if a link in the page hiphop.html needs to access the logo.png:

```
<a href="../images/logo.png"> Logo </a>
```



/EXERCISES

- To link the hiphop page from the music homepage:
`<a href="hiphop.html"> Hip hop </a>`
- To load the css file from the homepage in the root:
`<a href="css/styles.css"> CSS </a>`
- To link the camping file from the homepage in the root:
`<a href="activities/indoor/camping.html"> Camping </a>`
- To link the camping file from the outdoors homepage:
`<a href="../indoor/camping.html"> Camping </a>`
- To link to the homepage from the basketball page:
`<a href="../../index.html"> Homepage </a>`



<ANCHORS>

/JUMPING ANCHORS



/JUMPING ANCHORS

- An anchor link is a link, which allows the users to flow through a website page. It helps to scroll and skim-read easily

- Two simple steps:

1. Add an id to the element you want to add the anchor

```
<h2 id="anchor-name">Section name</h2>
```

2. Create a link that points to that id

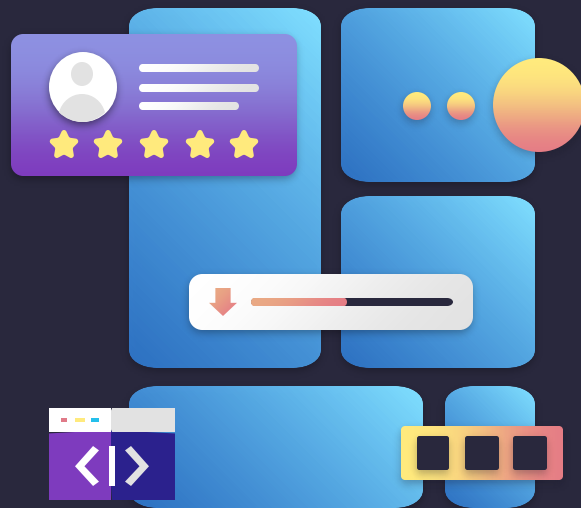
```
<a href="#anchor-name">Jump to the "anchor-name" id
```

```
</a>
```

- See the effect: <https://www.w3docs.com/tools/code-editor/1044>

/IMAGES

Creating links between pages, to
other sites and email links



/IMAGES

We will learn how to:

- Include images to websites
- Pick which image format to use
- Set image dimensions
- Optimize images for page rendering

A PICTURE IS WORTH A THOUSAND WORDS

[Sign up](#)



**/A PICTURE
IS WORTH A
THOUSAND
WORDS**

SIGN UP

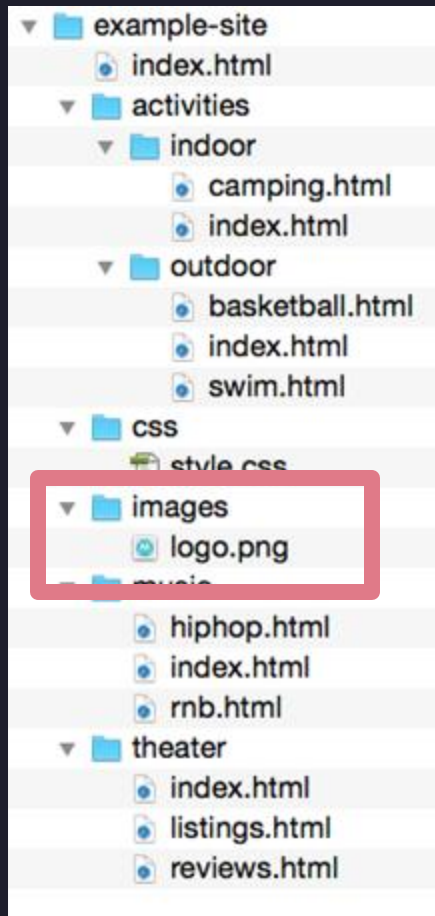
/HERO SECTION

A hero section is the area that immediately shows up on the screen under your logo and menu to show:

- What you have to offer
- Why people should trust you
- The benefits of working with you
- What action they should take next
- Your brand standing out from the competition

/STORING IMAGES

If you are building websites from scratch, it is a good practice to create a folder for all of the images the site uses.





ADDING IMAGES



TEXT DESCRIPTION
OF IMAGE IF IT
DOES NOT RENDER



```

```



TELLS THE
BROWSER
THE SOURCE
OF THE
IMAGE, FILE
LOCATION



TEXT
DESCRIPTION
DISPLAYED
WHEN USERS
HOVERS
OVER THE
IMAGE

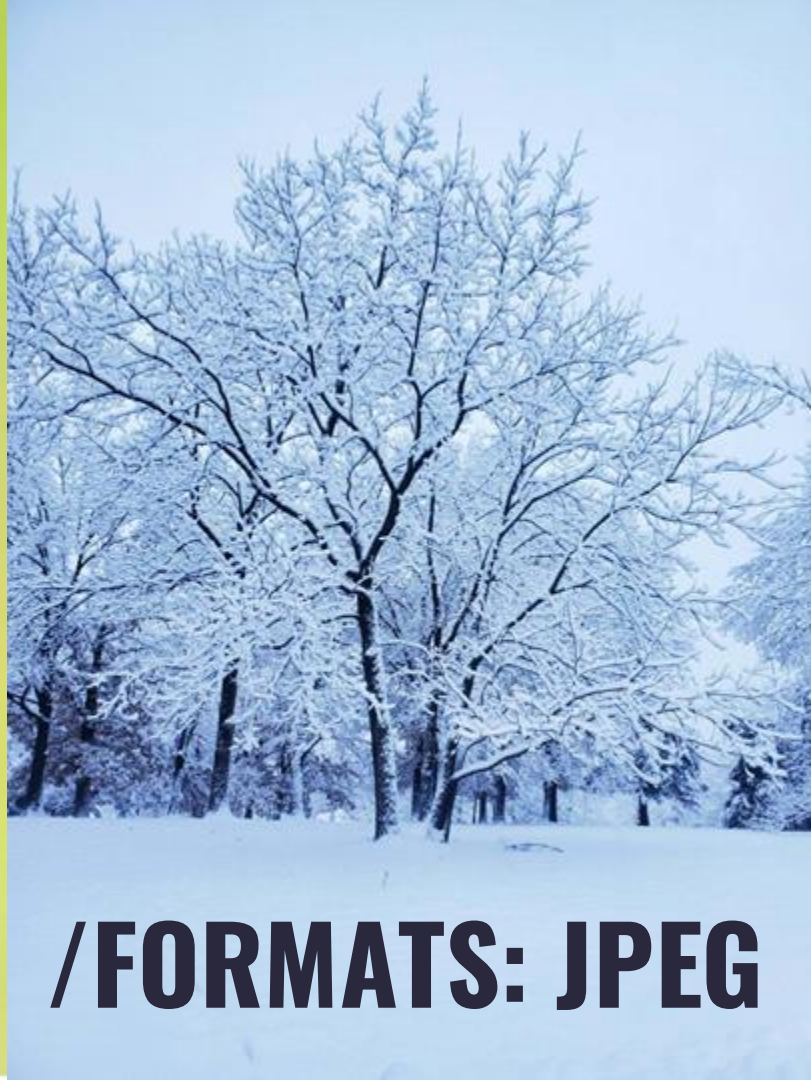
> HEIGHT & WIDTH OF IMAGES <

THIS SPECIFIES THE
WIDTH OF THE
IMAGE IN PIXELS

```

```

THIS
SPECIFIES
THE HEIGHT
OF THE
IMAGE IN
PIXELS



/FORMATS: JPEG



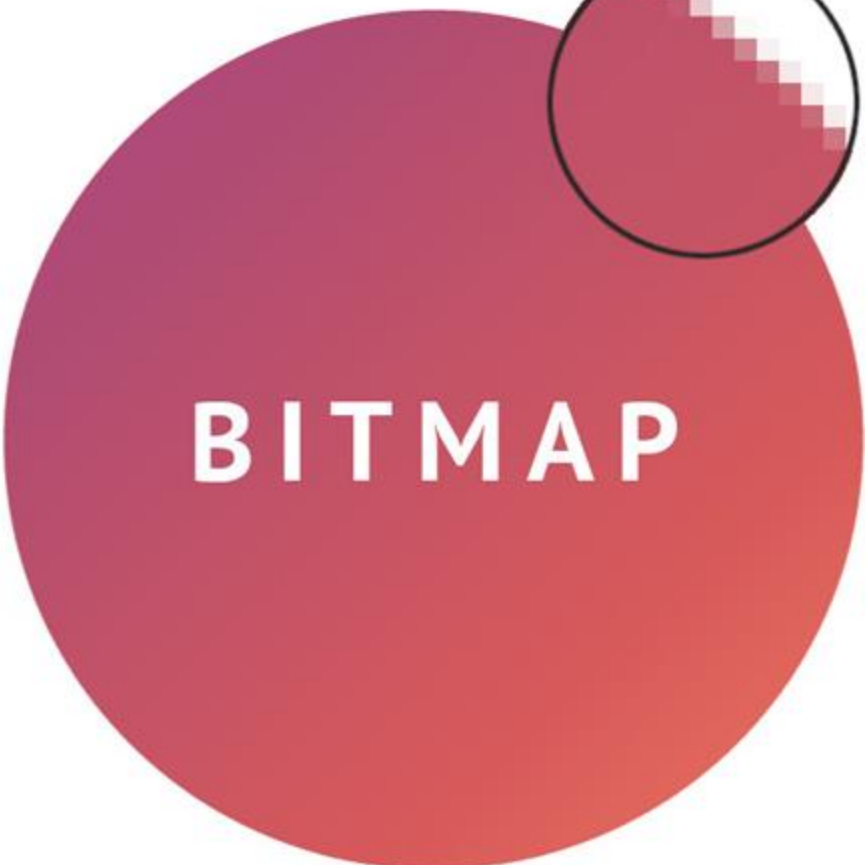
/FORMATS: PNG & GIFs



/FORMATS: SVG



VECTOR



BITMAP



/FORMATS: GIF

> HTML5: FIGURE AND FIGURE CAPTIONS <

```
<figure>
  
  <br/>
  <figcaption>
    Sea otters hold hands when they sleep so they don't
    drift away from each other.
  </figcaption>
</figure>
```



Sea otters hold hands when they sleep so they don't drift away from each other.

/DOWNLOAD

The `download` attribute specifies that the target (the file specified in the `href` attribute) will be downloaded when a user clicks on the hyperlink.

```
<a href="relative/path/to/file.txt" download>Download</a>
```

/COMMENTS

Web designed add comments to HTML for these reasons:

1. To add notes describing the purpose of an HTML element or block
2. To hide code that is unfinished
3. To hide code that is causes issues or brakes other parts of a page.
4. To indicate the beginning or ending of a large block



/ID AND CLASS

- Every HTML element can carry a unique `id` attribute. Its value should start with a letter or underscore.
- Every HTML element can carry a `class` attribute that can be used in multiple HTML elements.
- Both attributes are often used as selectors of HTML elements in CSS rules and JavaScript.

```
<p id="f50" class="footer">
```


/DIVS AND SPANS

- The `<div>` element allows you to group a set of elements together in one block-level box (section).
- The `<span>` element acts like an inline equivalent of the `<div>`.

```
<div class="footer">
```

```
    <p> Contact us <span class="bolder"> now! </span>
```

```
    </p>
```

```
</div>
```

/NAV

- The `<nav>` defines a set of navigation links.
- Notice that NOT all links of a document should be inside a `<nav>` element.
- The `<nav>` element is intended only for major blocks of navigation links.

```
<nav>
```

```
    <a href= "page1.html">Menu item 1</a>
```

```
    <a href= "page2.html">Menu item 2</a>
```

```
    <a href= "page3.html">Menu item 3</a>
```

```
</nav>
```

/HEAD TAG

- The `<head>` element is a container for the following elements:

`<title>`, `<style>`, `<meta>`, `<link>`, `<script>`, and `<base>`

```
<html>
```

```
<head>
```

```
...
```

```
</head>
```

```
<body>
```

/TITLE TAG

- The `<title>` tag defines the title of the document. The title must be text-only, and it is shown in the browser's title bar or in the page's tab.

```
<head>
```

```
    <title>Title of web page</title>
```

```
</head>
```

/FAVICON

- A favicon is a small image displayed next to the page title in the browser tab.

```
<head>  
  <title>Title of web page</title>  
  <link rel="icon" type="image/x-icon" href="/images/favicon.ico">  
</head>
```